

Compact USB-powered touch-activated lighting module. Includes TTP223 capacitive touch sensor, an NPN transistor and five white LEDs.

1 Features

- USB powered (5 V)
- Capacitive touch pad (TTP223) — no mechanical switch
- Five white LEDs with individual current-limiting resistors (150 Ω)
- NPN transistor (MMBT3904) as LED driver for stable switching
- Plug-and-play, no firmware required

2 Applications

- USB decorative lighting
- Touch technology demonstration
- Educational electronics projects
- DIY gadgets and prototypes

3 Description

The Capacitive Touch LED USB Module is a compact lighting device designed to demonstrate the use of capacitive touch technology in an accessible and visually attractive way. The board integrates the TTP223-BA6 capacitive touch sensor, which replaces traditional mechanical switches and provides reliable, contactless activation. A dedicated touch pad is printed on the PCB surface and connected directly to the sensor input.

When a user touches the pad, the sensor output drives an MMBT3904 NPN transistor, which in turn switches a group of five white surface-mount LEDs. Each LED is connected in series with an individual 150 Ω resistor, ensuring consistent brightness and current limiting. This design guarantees stable performance and protects the LEDs from overcurrent, while also improving durability and reliability.

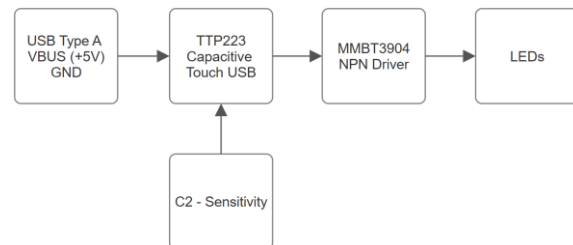
The module is powered directly from the USB VBUS (5V), using a standard Type-A male connector. The D+ and D- lines are not connected, as no communication with the USB host is required. The choice of a USB form factor makes the device plug-and-play, requiring no cables, external power supply, or programming. The total current consumption is typically under 100 mA, which allows the module to be safely

powered by any computer, charger, or power bank.

The overall design emphasizes simplicity, compactness, and usability. The USB stick shape makes the module portable and convenient, while the capacitive touch control gives it a modern and intuitive user interface. Additionally, the sensitivity of the touch pad can be easily adjusted by changing the value of the external capacitor (C2), allowing customization for different environments and applications.

This module can be used for educational demonstrations, DIY projects, or as a decorative USB lighting accessory. It also serves as a practical example of how to interface capacitive sensors with discrete electronics to control higher-current loads such as LEDs.

Block Diagram



PCB size (without components)

55mm X 18mm

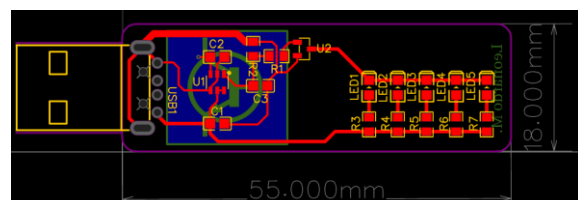


Figure 1. PCB Layout

Electrical Characteristics

Parameter	Min.	Typ.	Max.	Unit
Supply Voltage (VCC)	4.5	5.0	5.5	V
Operating Current	-	90	120	mA
Touch Detection Distance	-	0.5	2.0	mm
LED Forward Current (per LED)	-	10	17	mA

Pinout (USB connector)

Pin	Name	Description
1	VBUS	+5V from USB
2	D-	Not connected
3	D+	Not connected
4	GND	Ground

BOM (Bill of Materials)

Name	Designator	Quantity
USB-Type A Male	USB1	1
MMBT3904Q	U2	1
TTP223-BA6	U1	1
150R 0805 5%	R3, R4, R5, R6, R7	5
100k 0805 5%	R2	1
10k 0805 5%	R1	1
LED 0805 White	LED1, LED2, LED3, LED4, LED5	5
22pF 0805 X7R	C2	1
0.1uF 0805 X7R	C1, C3	2

Schematic

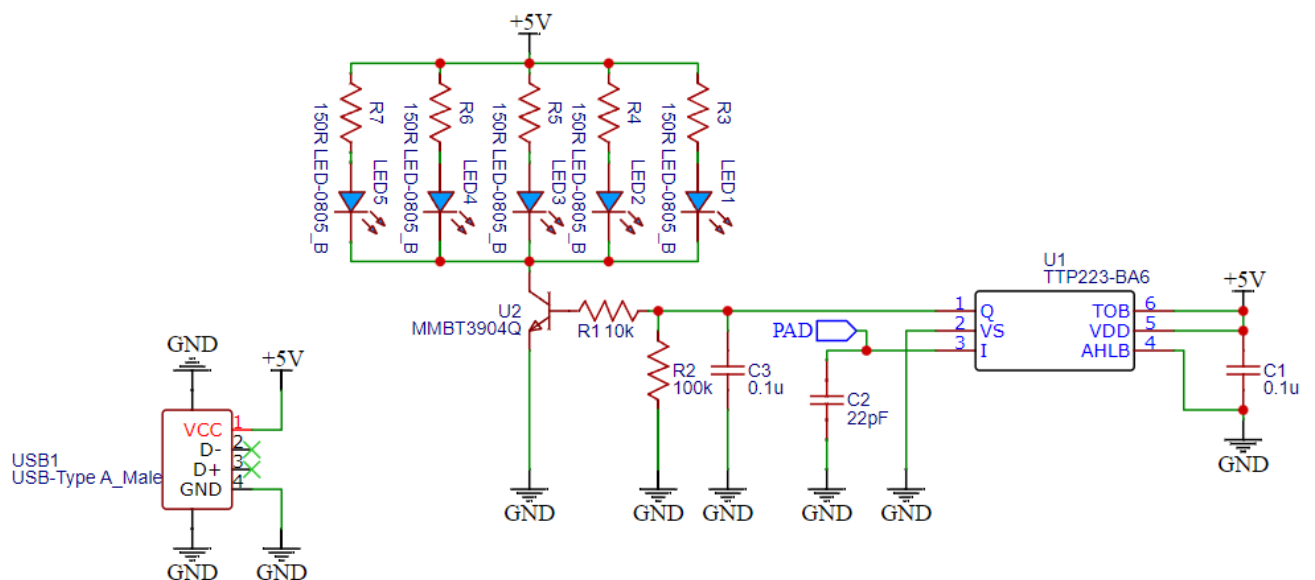


Figure 2. Schematic

3D image of the PCB

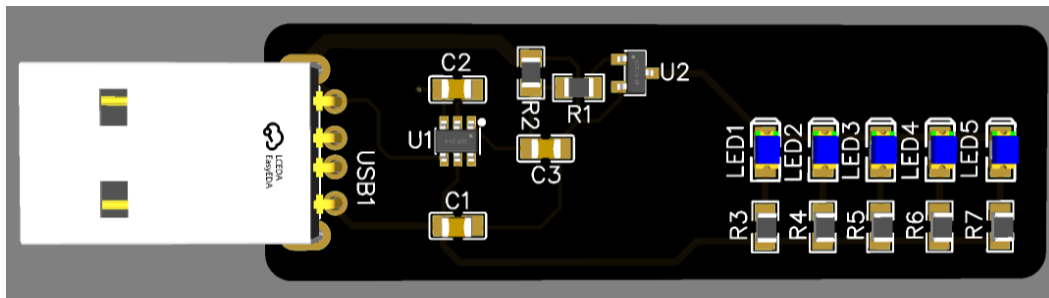


Figure 3. PCB Top Side



Figure 4. PCB Layout Bottom Side

PCB Assembled

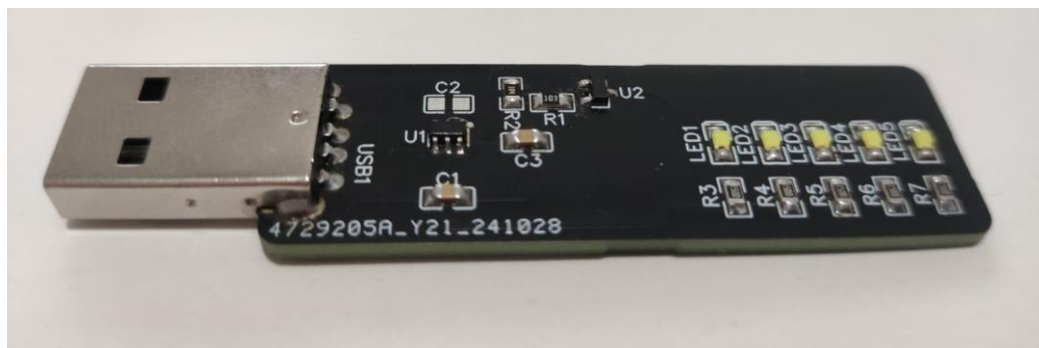


Figure 5. PCB Assembled

Disclaimer

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